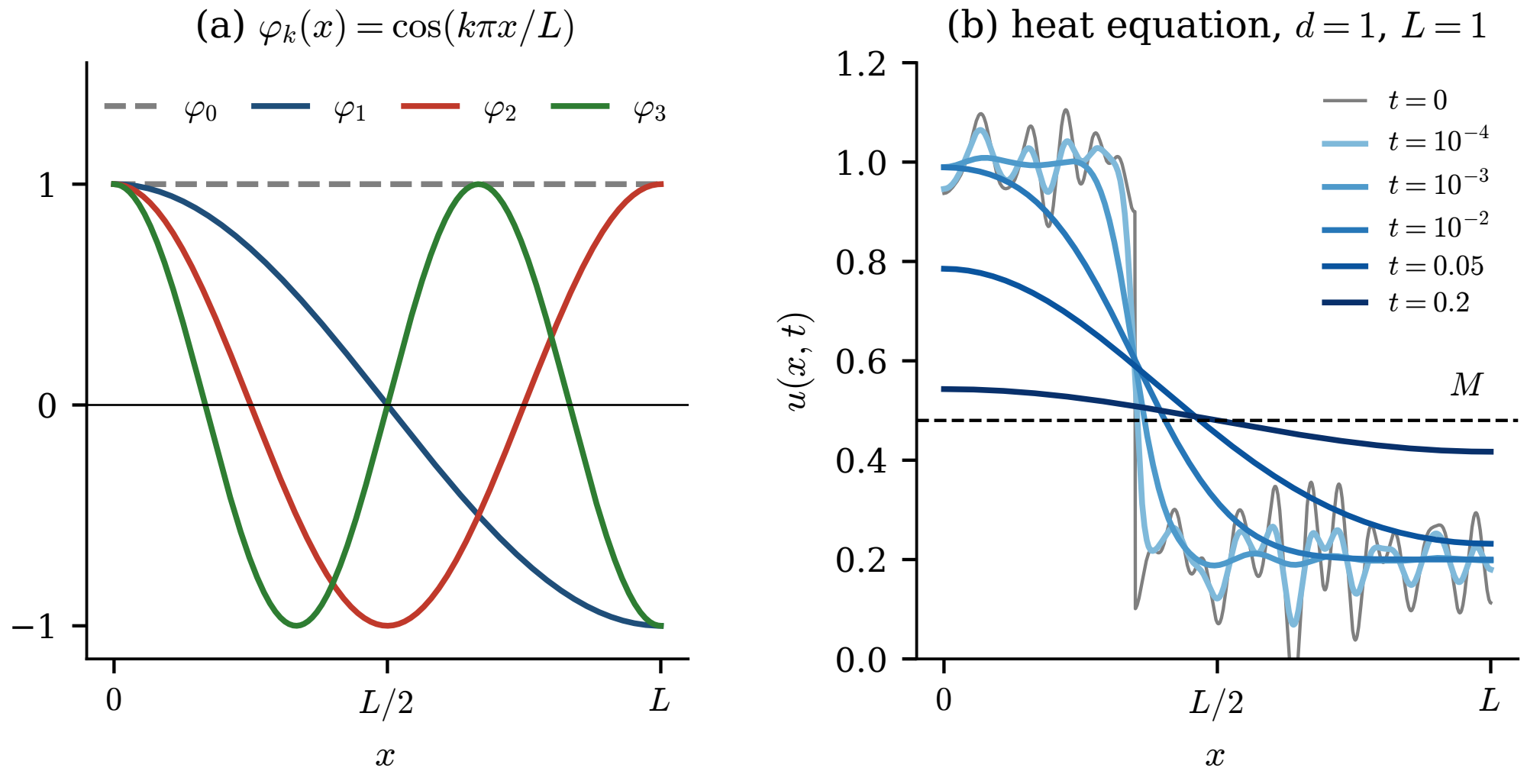
(a) $\varphi_k(x) = \cos(k\pi x/L)$
φ_0
φ_1
φ_2
φ_3
1
0
−1
0
$L/2$
L
x
(b) heat equation, $d = 1$, $L = 1$
1.2
1.0
0.8
0.6
0.4
0.2
0.0
$u(x, t)$
$t = 0$
$t = 10^{-4}$
$t = 10^{-3}$
$t = 10^{-2}$
$t = 0.05$
$t = 0.2$
M
0
$L/2$
L
x